Position Dependence of Phase-Decomposed Lag-One Covariance in Prime-Gap Data

A preregistered falsification and fixed-count empirical study

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Abstract

We report the falsification of a preregistered prediction concerning the phase-decomposed lag-one covariance of consecutive prime gaps, together with a subsequent exploratory diagnostic. Earlier work established persistent negative lag-one dependence and an exact decomposition of the covariance of locally detrended gaps into A + X + Y + D. For a primorial modulus M, we define the non-D covariance share S(M) = (A + X + Y) / Cov(e_n, e_{n+1}). A preregistered prediction that S(M) would not decrease at higher window positions was falsified: across three windows from 10^8 to 3×10^10, S(M) decreased for every tested modulus. In an exploratory fixed-count diagnostic using eight windows with approximately 10^8 gaps each, spanning prime locations from 10^8 to 10^13, both S(M) and |ρ_resid(M)| decreased monotonically with L = log p. The fixed-count design strongly disfavors a pure sample-size explanation. Over L ∈ [19.96, 29.93], logarithmic and power-law descriptions are similarly supported, while the reciprocal form is less supported at large moduli (Table A.6); the linear-in-L form is substantially disfavored. The results are reported as finite-range empirical trends, without an asymptotic interpretation or proposed theoretical mechanism.

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1. Introduction

1.1 Background

Papers 1 and 2 established two empirical facts about the lag-one covariance of consecutive prime gaps g_n = p_{n+1} − p_n.

Paper 1 (Zenodo DOI 10.5281/zenodo.23162269) reported persistent negative lag-one dependence in locally detrended prime-gap windows up to $N = 10^9$. The elementary independent-gap version of the Cramér heuristic does not encode the observed negative lag-one dependence [3, 4].

Paper 2 (Zenodo DOI 10.5281/zenodo.23222536) introduced an ABCD decomposition

of the lag-one covariance. Let $e_n = g_n - (\alpha + \beta \cdot \log p_n)$ denote the residual after the global detrending $g \approx \alpha + \beta \cdot \log p$ used throughout this work. For each primorial modulus M, the residual sequence is partitioned by the opening-prime residue $r_n = p_n \bmod M$, and the residual is written as $e_n = \mu(r_n) + u_n$, where $\mu(r)$ is the phase mean for residue r and u_n is the within-class residual. The lag-one covariance of the detrended gaps then decomposes exactly as $\mathrm{Cov}(e_n, e_{n+1}) = A + X + Y + D$, where $A = E[\mu_{prev} \cdot \mu_{cur}]$, $X = E[\mu_{prev} \cdot u_{cur}]$, $Y = E[u_{prev} \cdot \mu_{cur}]$, $D = E[u_{prev} \cdot u_{cur}]$. The non-D covariance share is $S(M) = (A + X + Y) / \mathrm{Cov}(e_n, e_{n+1}) = 1 - D / \mathrm{Cov}(e_n, e_{n+1})$, an algebraic ratio of the non-D covariance components to the total; it is not a causal 'fraction explained' and may exceed one or be negative if the cross terms X or Y dominate. In the single window $[10^{10}, 3\times10^{10})$ with 844,953,415 gaps, S(M) ranged from approximately 1.5% at M = 6 to 60.4% at M = 9699690. The residual correlation $\rho_{resid}(M) = D / \mathrm{var}_{within}$ was strictly negative at every modulus tested.

1.2 The preregistered question

The present work asked whether the phase-decomposed covariance structure changes across a sequence of logarithmic windows at different prime locations. The preregistered question was whether the non-D share S(M) would be nondecreasing as the window moved upward on the prime axis.

A protocol (paper3_protocol_v1.md) was committed to the source repository before any computation (commit 13d16db, 2026-10-12; file SHA-256 in the data inventory). No third-party timestamp registry was used; the hash establishes file integrity, not the timing of its conception. It specified seven predictions T1–T7 to be tested on a fixed set of three windows: $W1 = [10^8, 3\times10^8)$, $W2 = [10^9, 3\times10^9)$, $W3 = [10^{10}, 3\times10^{10})$ (the window of Paper 2). The three windows span a factor of approximately 300 in prime location p (from 10^8 to 3×10^{10}), corresponding to a factor of approximately 1.24 in $L = \log p$ (from $L \approx 18.97$ in W1 to $L \approx 23.57$ in W3). The protocol contained an abort condition: if any of T1–T7 failed, computation was to stop and the failure reported without modification of the predictions.

1.3 What happened

The algebraic checks T1–T2, the within-window empirical patterns T3–T5, and the W3 reproduction check were successful. T6 failed. T7 was descriptive because protocol v1 did not specify a numerical smoothness criterion.

T6 predicted that S(M) would be greater at higher windows than at lower windows, for every modulus M. The data falsify this prediction for every modulus tested: S(M) decreases monotonically from W1 to W3.

The failure motivated — not predetermined — the exploratory fixed-count diagnostic reported in Section 4. Across eight windows with approximately 10^8 gaps each, spanning prime locations from 10^8 to 10^{13}, S(M) and $|\rho_{resid}(M)|$ both decrease monotonically with $L = \log p$ for every M, while the gap count varies by 5.1%. A four-model comparison shows that the functional form of this dependence is not identified by the available data.

1.4 Structure of this paper

Section 2 describes the methods inherited from Papers 1–2 and the modifications introduced in Paper 3. Section 3 reports the preregistered tests and the falsification of T6. Sections 4–6 are explicitly exploratory and are distinguished from the preregistered analysis of Section 3: Section 4 describes the fixed-count diagnostic designed to separate gap count from window position, Section 5 quantifies the position dependence, and Section 6 compares candidate functional forms. Section 7 states the paper's claims and non-claims. Section 8 discusses open questions.

2. Methods

2.1 Overview

The methods of this paper are inherited from Papers 1 and 2 with three documented additions: runtime window arguments, a dynamic modulus-eligibility rule, and a model-comparison layer. This section describes the shared computational pipeline, the definitions used throughout the paper, and the specific additions. The source code, binary, and data files are recorded with their SHA-256 hashes in the data inventory (paper3_data_inventory.txt).

2.2 Prime-gap enumeration

Prime gaps are enumerated by an odd-only segmented sieve of Eratosthenes over the interval [LO, HI). The sieve uses a base-prime set up to ⌈√HI⌉ + 100. Segments contain 2 × 50,000,000 odd candidates each (approximately 10^8 integers per segment). For each window, we record the sequence g_n = p_{n+1} − p_n, where p_n is the n-th prime in [LO, HI) and p_{n+1} is the next prime, possibly greater than HI. The first and last opening primes, the minimum and maximum gap, and the total gap count n are recorded for each window.

The sieve is validated on two small windows, [10^6, 10^7) and [10^7, 10^8), where the counted numbers of primes match the values 586,081 and 5,096,876 exactly [7, 8] (derived from known prime-counting data). The same prime-generation pipeline was used in Paper 2 and reproduces the Paper 2 result at W3 (Section 3.5). These checks verify the implementation at small scale and reproduce the Paper 2 computation at 10^{10}. An independent enumeration check at 10^{12} is reported in Appendix B; a second independent check near 10^{13} remains desirable but is not performed here.

2.3 Window definitions

Three window families are used.

Preregistered windows (Section 3). Three windows of the form [10^k, 3 × 10^k) for k = 8, 9, 10: W1 = [10^8, 3×10^8), W2 = [10^9, 3×10^9), W3 = [10^{10}, 3×10^{10}). W3 is the window analysed in Paper 2.

Fixed-count windows (Sections 4–6). Eight windows, each calibrated to yield approximately 10^8 gaps, spanning prime locations from 10^8 to 10^{13}: A, P1, B, P2, P3,

C, P4, P5 with parameters listed in Table 4.1. Gap counts range from 95.98×10^6 to 100.85×10^6, a spread of 5.1%. Window A and window P5 differ by a factor of approximately 1.5 in L = log p.

Window parameters are supplied at runtime. In the source used for Papers 1–2, the window bounds LO and HI were compile-time constants. The present paper modifies the linear-analysis executable so that LO and HI are passed as command-line arguments, permitting any specified window to be analysed without recompilation. This is a code-level change; it does not affect the arithmetic.

2.4 Detrending

For each window we fit a global trend $g \approx \alpha + \beta \cdot \log p$ by ordinary least squares over the full window. The residual sequence is $e_n = g_n - (\alpha + \beta \cdot \log p_n)$. The linear form is retained as the primary detrending throughout Sections 3–6.

A quadratic diagnostic is also used as a robustness check. For the quadratic fit, we use $g \approx a + b \cdot z + c \cdot z^2$, $z = \log p - 0.5 \cdot (\log LO + \log HI)$, so that z is centred within the window. The quadratic diagnostic is reported with the exploratory robustness checks and is referenced again among the paper's limitations and claims.

The symbol u_n denotes the within-class residual throughout; the centered detrending variable in §2.4 and Appendix C is written z to avoid collision.

The numerical stability of the detrending is verified explicitly for the four narrowest fixed-count windows. The raw variance of log p varies from approximately 0.54 in the lowest-position window (A) to approximately 10^-8 in the highest (P5). The OLS slope β is therefore imprecise at high position; its standard error reaches ~31 in P5. We verified by running an independent implementation (Appendix C) with a centered variable z = log1p((p − p_mid)/p_mid) that the reported S(M) and ρ_resid(M) are identical to six decimal places in the four narrow windows tested (P3, P4, P5, C) and at every modulus in those windows, and that the residual variance Var(e) is also unchanged.

2.5 ABCD decomposition

For a modulus M, let $r_n = p_n \bmod M$ denote the residue of the opening prime. Let $\mu(r)$ denote the mean of e_n over all n in the window with $r_n = r$, and let $u_n = e_n - \mu(r_n)$ denote the within-class residual. Then the lag-one covariance of the detrended gaps decomposes exactly as $\mathrm{Cov}(e_n, e_{n+1}) = A + X + Y + D$. Covariance here denotes the lag-one second moment $(1/(N-1)) \Sigma\, e_n\, e_{n+1}$ over pairs with both indices in the window; it is not the fully centered sample covariance, since the two endpoint residuals do not appear in both roles. The distinction contributes at the 10^-13 level for windows of the size used here. The four components are defined as $A(M) = E[\mu(r_n) \cdot \mu(r_{n+1})]$, $X(M) = E[\mu(r_n) \cdot u_{n+1}]$, $Y(M) = E[u_n \cdot \mu(r_{n+1})]$, $D(M) = E[u_n \cdot u_{n+1}]$. The expectation is taken over pairs (n, n+1) in the window. The final gap remains in one-point quantities such as class means and variances; only the final lag-one pair is absent from covariance and ABCD sums because e_{n+1} is not available within the analysed gap sequence.

Two derived quantities are used throughout: $S(M) = (A(M) + X(M) + Y(M)) / \mathrm{Cov}(e_n,$

e_{n+1}) = 1 − D(M) / Cov(e_n, e_{n+1}) (non-D covariance share), ρ_resid(M) = D(M) / var_within (residual correlation), where var_within is the empirical within-class second moment of the residuals, equal to the within-class variance under the sample centring used here.

The ABCD decomposition closes to machine precision in every window computed for this paper; the maximum absolute deviation $|A + X + Y + D - \mathrm{Cov}(e_n, e_{n+1})|$ is reported per window and never exceeds 7.1×10^{-14}.

2.6 Modulus eligibility

The candidate ladder is $M \in \{6, 30, 210, 2310, 30030, 510510, 9699690\}$. After Pass 1, we compute the smallest non-empty residue-class count for each candidate modulus. A modulus is retained as standard when this count is at least 100, retained as diagnostic when it lies between 50 and 99, and excluded when it is below 50.

This rule was introduced for the multi-window extension after the small W1 window revealed that the largest candidate modulus could contain sparse classes. The preregistered W1–W3 analysis was initiated with the fixed candidate ladder of protocol v1. After the W1 sparsity issue was observed, the minimum-class eligibility rule was documented and used for the multi-window extension. The rule does not affect the seven retained W1–W3 moduli used in Section 3, though $M = 510510$ is flagged as diagnostic in W1 (min_class = 79), nor the W3 reproduction of Paper 2 at $M = 9699690$.

Relative to the Paper 2 statistical calculation, the minimum-class eligibility rule is the only change to the set of moduli entering a window-level analysis. Runtime arguments, expanded diagnostics, and output handling are implementation changes; the prime-generation, detrending, and ABCD formulas are otherwise inherited.

2.7 Bootstrap

Within-window uncertainty in ρ_resid is estimated by a non-overlapping block bootstrap based on 40 contiguous blocks per window [5, 6]. Each replicate samples 40 of these pre-defined blocks with replacement and concatenates their within-block and boundary contributions. For each modulus M, 1000 replicate resamples are drawn. The seed base for the random number generator is 20261006, incremented per modulus as $20261006 + k \cdot 1000$.

The bootstrap confidence intervals reported throughout this paper are marginal within-window intervals. They are not used to assign a joint probability to any cross-window pattern, and they are not treated as independent tests across moduli, because the values of ρ_resid across M within a single window are derived from the same gap sequence and are strongly dependent. Confidence intervals for the fitted exponents α(M) of Section 5.4 are not computed; those values are reported as descriptive summaries only.

2.8 Computational structure

Each window is analysed in two passes over the gap sequence.

Pass 1 accumulates the ordinary-least-squares moments for the detrending fit, and the per-residue sums of g_n and log p_n for each modulus. It produces the OLS coefficients α and β, the class means μ(r), and the minimum class count used for eligibility.

Pass 2 re-streams the gap sequence and accumulates, for each modulus, the four ABCD components, the residual variance, the within-class variance, and the block-level quantities used in the bootstrap. The final covariance, S(M), and ρ_resid(M) are computed at the end of Pass 2 from the accumulated sums.

The two-pass structure avoids storing the full gap sequence. Memory usage is independent of the number of gaps, apart from fixed arrays indexed by the selected moduli and by the 40 bootstrap blocks. The largest window analysed in this paper has 844,953,415 gaps.

2.9 Model comparison

Section 6 compares four two-parameter models for S(M, L) and |ρ_resid(M, L)|:

Label	Form	Parameters
power	y = a · L^(−α)	(a, α)
reciprocal	y = a + b / L	(a, b)
log	y = a + b · log(L)	(a, b)
linear	y = a + b · L	(a, b)

All four have two free parameters. Fits use nonlinear least squares with moment-based starting values. Two metrics are reported in the main text: the residual sum of squares from the full fit, and the small-sample corrected Akaike Information Criterion AICc = n · ln(RSS / n) + 2k + 2k(k + 1) / (n − k − 1). All comparisons are made on the same data (fixed-count set or combined set) and use the same metric definitions. Because n = 8 for the fixed-count set, AICc rankings are treated as descriptive sensitivity summaries rather than decisive model-selection evidence (Section 6.2).

2.10 Data and code

The linear-analysis executable test_multiwindow_v1.exe performs the pipeline of Sections 2.2–2.8 for any window passed as LO HI arguments. A separate executable test_multiwindow_quad_v1.exe performs the quadratic diagnostic of Section 2.4. The Python script paper3_model_comparison.py performs the model comparison of Section 6. All source files, binaries, and CSV outputs are recorded with their SHA-256 hashes in paper3_data_inventory.txt.

The seed base for the bootstrap is 20261006 in every source. Any value of 20260106 appearing in earlier drafts is a typographical error.

3. Preregistered Tests and Falsification of T6

3.1 Protocol v1

Prior to any computation, a protocol (paper3_protocol_v1.md, 2026-10-09) specified seven predictions T1–T7 to be tested on three windows:

Window	Range	n_gaps	L = log p center
W1	$[10^8, 3\times10^8)$	10,490,870	18.9700
W2	$[10^9, 3\times10^9)$	93,602,003	21.2726
W3	$[10^{10}, 3\times10^{10})$	844,953,415	23.5752

The protocol contained an abort condition: if any of T1–T7 failed, computation was to stop and the failure reported without modification of the predictions.

3.2 Methods Used in the Preregistered Tests

For each window we computed the lag-one covariance Cov(e_n, e_{n+1}) of consecutive prime gaps g_n = p_{n+1} − p_n with opening prime p_n ∈ [LO, HI), a global detrending g ≈ α + β·log p fitted by OLS, giving e_n = g_n − (α + β · log p_n), per-residue-class means μ[r] of residuals for each M ∈ {6, 30, 210, 2310, 30030, 510510, 9699690} with r = p mod M, and the ABCD decomposition

Cov(e_n, e_{n+1}) = A + X + Y + D,

where A = E[μ_prev · μ_cur], X = E[μ_prev · u_{n+1}], Y = E[u_n · μ_cur], D = E[u_n · u_{n+1}]. The non-D covariance share is S(M) = (A + X + Y) / Cov, and the residual correlation is ρ_resid(M) = D / var_within. Block bootstrap confidence intervals used 40 blocks, 1000 replicates, and seed base 20261006 + k·1000.

Modulus eligibility was determined dynamically by the minimum class count over all residues, with a modulus excluded when its smallest non-empty class count is below 50 and classes between 50 and 99 flagged diagnostic. This rule replaced the fixed seven-modulus list that had been used in Paper 2, and constitutes the only logical change to the inherited pipeline.

3.3 Results

Table 3.1 — Preregistered test outcomes.

ID	Prediction	Outcome	Evidence
T1	$\lvert A+X+Y+D - Cov \rvert < 10^{-6}$ every row	PASS	max deviation 7.1×10^{-15} (W1), 7.1×10^{-15} (W2), 7.1×10^{-14} (W3)
T2	var_between + var_within = var_e every row	PASS	max deviation ≤ 5.7×10^{-14}
T3	S(M) monotone increasing in M each window	PASS	monotone in all three windows (6, 6, 7 moduli)
T4	ρ_resid < 0 for every M in every window	PASS	19/19 values negative
T5	Bootstrap 95% CIs exclude zero	PASS	19/19 intervals strictly negative
T6	S(M) at higher window ≥ S(M) at lower window	FAIL	all adjacent window pairs decrease, and all 18 pairwise window–modulus comparisons decrease
T7	A, X, Y, D vary smoothly across windows	Descriptive; not formally testable as preregistered	see §3.3.1

3.3.1

Protocol v1 did not define a numerical smoothness criterion for T7. The A, X, Y, D components show no visually apparent discontinuity across W1–W3, but T7 is reported descriptively rather than scored as pass or fail.

3.4 Falsification of T6

T6 predicted that S(M) would be greater in windows at higher position than at lower position, for every M. The data falsify this in every case.

Table 3.2 — S(M) across W1, W2, W3.

M	S(W1)	S(W2)	S(W3)	W1→W3 decrease
6	0.019670	0.017458	0.015317	22.1%
30	0.162635	0.142455	0.125040	23.1%
210	0.323706	0.296759	0.272725	15.7%
2310	0.437143	0.410226	0.385438	11.8%
30030	0.525091	0.500907	0.475868	9.4%
510510	0.592241	0.568517	0.544106	8.1%
9699690	—	—	0.604431	(n/a)

S(M) decreases monotonically from W1 to W3 for every modulus tested. Across the six moduli common to W1–W3, the W1-to-W3 decrease ranges from approximately 8% at M = 510510 to 23% at M = 30. The sign of the change is opposite to the prediction: all adjacent window pairs decrease, and all 18 pairwise window–modulus comparisons decrease.

W1, W2, W3 differ simultaneously in gap count n and in window position L = log p. The falsification of T6 does not by itself distinguish between these two explanatory variables. Section 4 addresses this ambiguity with a fixed-count diagnostic.

3.5 Reproduction of Paper 2

The window W3 = $[10^{10}, 3\times10^{10})$ is the window used in Paper 2. Re-running the v1 pipeline on W3 reproduces the central result of Paper 2:

Quantity at M = 9699690	Paper 2	This work (W3)
S(M)	60.4%	60.4431%
ρ_resid	−0.010306	−0.010306
|ABCD − Cov|	$< 10^{-6}$	$< 10^{-13}$

This confirms that the prime-generation, detrending, and ABCD calculation reproduce the Paper 2 result at W3. Paper 3 additionally applies a documented post-Pass-1 modulus-eligibility rule based on the observed minimum class count; this rule does not affect the W3 M=9699690 calculation.

3.6 What the Falsification Does and Does Not Establish

The failure of T6 is decisive as a statistical fact: across three windows spanning a factor of approximately 300 in prime location (p from 10^8 to 3×10^{10}), S(M) decreases monotonically with position for every modulus tested. Bootstrap intervals quantifying within-window uncertainty in ρ_resid are reported for the fixed-count set, but S(M) itself does not carry a bootstrap interval.

The falsification does not identify the cause of the decrease. Two candidate explanations remain open:

- **H1 (size):** S depends on the number of gaps n.
- **H2 (position):** S depends on the location of the window on the prime axis, represented by L = log p.

W1, W2, W3 vary both quantities together, so H1 and H2 are not distinguishable from Table 3.2 alone. Section 4 reports a diagnostic designed to separate them.

3.7 Protocol Abort Clause

Protocol v1 specified that if any of T1–T7 failed, computation would stop. In accordance with this clause, the execution of the original phase order (W1 → W2 → W3 → W4 → W5 → W6) was halted after W3.

The follow-up diagnostics in Sections 4–6 were conducted under a separate exploratory protocol (paper3_protocol_v2.md). Protocol v2 was committed to the source repository after the diagnostics were run (commit b143b51, 2026-10-12; file SHA-256 in the data inventory); the diagnostics are therefore exploratory, not preregistered. No third-party timestamp registry was used; the hash establishes file integrity, not the timing of its conception.

4. Fixed-Count Diagnostic

4.1 Motivation

Section 3 established that S(M) decreases from W1 to W3 for every M, but W1, W2, W3 differ simultaneously in gap count n and in position L = log p. To distinguish between H1 (size) and H2 (position), we designed a set of windows in which n is held approximately fixed and L is varied.

If S depends on n, it should remain approximately constant across this set. If S depends on position, it should vary.

4.2 Window Design

Eight windows were constructed, each calibrated to yield approximately 10^8 gaps. The five windows P1–P5 were designed first; the three auxiliary points A, B, C were added subsequently: A extends the range toward lower position, while B and C were placed at nearly the same positions as P2 and P4 respectively to provide an internal reproducibility check, giving eight fixed-count windows in total.

Table 4.1 — Fixed-count window set.

Label	Range	n_gaps	L = log p center
A	$[10^8, 2.18\times10^9)$	100,849,143	19.9616
P1	$[10^9, 3.145\times10^9)$	100,239,099	21.2962
B	$[10^{10}, 1.222\times10^{10})$	95,979,254	23.1261
P2	$[10^{10}, 1.2305\times10^{10})$	99,639,099	23.1296
P3	$[10^{11}, 1.02535\times10^{11})$	100,039,703	25.3410
C	$[10^{12}, 1.00277\times10^{12})$	100,245,924	27.6324

Label	Range	n_gaps	L = log p center
P4	$[10^{12}, 1.002765\times10^{12})$	100,064,944	27.6324
P5	$[10^{13}, 1.0002995\times10^{13})$	100,055,061	29.9338

The gap count varies between 95.98×10^{6} and 100.85×10^{6}, a spread of 5.1%. The window set spans five decades in prime location p (10^{8} to 10^{13}), corresponding to a factor of approximately 1.50 in L = log p (from 19.96 to 29.93).

Two pairs of windows were placed at nearly identical positions as an internal reproducibility check: (B, P2) at L ≈ 23.13 and (C, P4) at L ≈ 27.63.

4.3 S(M) Across the Fixed-Count Set

Table 4.2a — S(M) across the fixed-count set (M = 6, 30, 210).

Window	L	S(6)	S(30)	S(210)
A	19.9616	0.018154	0.148372	0.304672
P1	21.2962	0.017385	0.142180	0.296332
B	23.1261	0.015694	0.127535	0.275772
P2	23.1296	0.015679	0.127509	0.275648
P3	25.3410	0.014028	0.113996	0.255664
C	27.6324	0.012484	0.102751	0.240956
P4	27.6324	0.012483	0.102733	0.240951
P5	29.9338	0.011521	0.092723	0.226046

Table 4.2b — S(M) across the fixed-count set (M = 2310, 30030, 510510).

Window	L	S(2310)	S(30030)	S(510510)
A	19.9616	0.418677	0.509198	0.576448
P1	21.2962	0.409693	0.500101	0.567502
B	23.1261	0.387465	0.477681	0.546156
P2	23.1296	0.387432	0.477455	0.545785
P3	25.3410	0.366755	0.456336	0.525676
C	27.6324	0.351249	0.442187	0.511301
P4	27.6324	0.351294	0.442255	0.511412
P5	29.9338	0.334435	0.423067	0.494000

For every M, S decreases monotonically with L across the eight windows. The relative decrease from A to P5 (ΔL = 9.9722) is:

M	S(A)	S(P5)	Relative decrease
6	0.018154	0.011521	36.5%
30	0.148372	0.092723	37.5%

M	S(A)	S(P5)	Relative decrease
210	0.304672	0.226046	25.8%
2310	0.418677	0.334435	20.1%
30030	0.509198	0.423067	16.9%
510510	0.576448	0.494000	14.3%

The relative decrease varies with M, from 14% at M = 510510 to 37% at M = 30.

4.4 ρ_resid(M) Across the Fixed-Count Set

Table 4.3a — ρ_resid(M) across the fixed-count set (M = 6, 30, 210).

Window	L	ρ(6)	ρ(30)	ρ(210)
A	19.9616	−0.029194	−0.025414	−0.020913
P1	21.2962	−0.027907	−0.024444	−0.020197
B	23.1261	−0.025513	−0.022675	−0.018940
P2	23.1296	−0.025530	−0.022691	−0.018956
P3	25.3410	−0.022949	−0.020667	−0.017453
C	27.6324	−0.020527	−0.018683	−0.015875
P4	27.6324	−0.020526	−0.018682	−0.015874
P5	29.9338	−0.018701	−0.017189	−0.014718

Table 4.3b — ρ_resid(M) across the fixed-count set (M = 2310, 30030, 510510).

Window	L	ρ(2310)	ρ(30030)	ρ(510510)
A	19.9616	−0.017636	−0.015026	−0.013090
P1	21.2962	−0.017080	−0.014589	−0.012734
B	23.1261	−0.016130	−0.013857	−0.012136
P2	23.1296	−0.016142	−0.013873	−0.012153
P3	25.3410	−0.014935	−0.012904	−0.011335
C	27.6324	−0.013636	−0.011788	−0.010390
P4	27.6324	−0.013634	−0.011786	−0.010386
P5	29.9338	−0.012711	−0.011071	−0.009762

All 48 values are strictly negative. All 48 marginal bootstrap intervals within windows exclude zero. These intervals support the persistence of negative residual correlation within the analysed windows; they are not treated as 48 independent tests, since the values across M within a single window are derived from the same gap sequence and are strongly dependent.

The magnitude of ρ_resid decreases monotonically with L for every M, mirroring the behaviour of S.

4.5 Internal Reproducibility Check

Two pairs of windows lie at nearly identical positions: B (L = 23.1261) and P2 (L = 23.1296), separated by ΔL = 0.0035; and C (L = 27.6324) and P4 (L = 27.6324), with identical L.

Table 4.4 — Reproducibility at duplicated positions.

Pair	ΔL	$\lvert\Delta S(6)\rvert$	$\lvert\Delta S(510510)\rvert$	$\lvert\Delta\rho(6)\rvert$	$\lvert\Delta\rho(510510)\rvert$
B vs P2	0.0035	1.5×10^{-5}	3.7×10^{-4}	1.7×10^{-5}	1.7×10^{-5}
C vs P4	0.0000	1.0×10^{-6}	1.1×10^{-4}	1.0×10^{-6}	4.0×10^{-6}

The differences between the paired estimates are ≤ 3.7×10^-4 in S and ≤ 1.7×10^-5 in ρ. Since the tables report six decimal places, these differences exceed the last printed digit for the larger-modulus S values, but they remain roughly 200 times smaller than the total A-to-P5 decrease reported in Section 5. Because the paired windows overlap substantially (B is a strict subset of P2, with ~96% of P2's gap count shared; C contains P4, with ~99.8% of C's gap count shared with P4), the numerical agreement reported here is not an independent replication. It demonstrates insensitivity of the estimates to small changes in the window endpoint at essentially fixed position. It does not rule out a systematic error shared by all runs.

4.6 Interpretation

Under the constraint n ≈ 10^8 ± 5.1%, S and |ρ_resid| both decrease monotonically with L for every M tested. Because n varies by less than 5.1% while S varies by 14–37%, the fixed-count diagnostic strongly disfavours a pure sample-size explanation. The design does not establish a unique causal variable; see §4.7.

H1 (pure sample-size explanation) is strongly disfavoured by this diagnostic, given the approximately fixed n and the smooth, monotone decline of S across positions. A quantitative exclusion of every conceivable sample-size mechanism would require sampling multiple n values at each position; this is not attempted here.

4.7 Caveats

Three limitations must be stated explicitly.

(a) The gap count is approximately, not exactly, fixed. n varies from 95.98×10^6 to 100.85×10^6, a spread of 5.1%. This is small compared to the 14–37% variation in S, but it is not zero.

(b) Window width in x varies. Holding n approximately fixed requires widening the window as position increases, since prime density decreases. Window A spans 2.08×10^9 in x; window P5 spans 2.995×10^9 in x. The width in x varies only by a factor of about 1.44 across the eight windows; the quantity that varies strongly is the relative span Δx/x, which falls from ~20.8 in window A to ~3×10^-4 in P5.

(c) Position is represented by L = log p. We use L as a proxy for window location. This is the natural choice given the heuristic scale of prime gaps, but it does not exclude

the possibility that the true explanatory variable is some other monotone function of position.

An additional note: the fixed-count set and the variable-count set of Section 3 are not directly comparable. They differ in design and should not be pooled without accounting for this difference; the model comparison of Section 6 is presented separately for this reason.

4.8 Summary

Within eight fixed-count windows spanning L ∈ [19.96, 29.93] at n ≈ 10^8:

- S(M) decreases monotonically with L for every M tested.
- ρ_resid(M) decreases in magnitude monotonically with L for every M tested.
- The total relative decrease is 14–37% depending on M.
- The paired estimates differ by at most 3.7 × 10^-4 in S (at M = 510510) and 1.7 × 10^-5 in ρ_resid. These differences exceed the last printed digit for the larger-modulus S values but remain ~200× smaller than the total A-to-P5 decrease reported in Section 5.
- The fixed-count diagnostic strongly disfavours a pure sample-size explanation. The functional form of the position dependence is examined in Section 5.

5. Empirical Position Dependence

5.1 Scope

This section quantifies the position dependence established in Section 4. Three questions are addressed: how much S(M, L) and ρ_resid(M, L) decrease across the observed range; whether the rate of decrease depends on M; and whether any simple rescaling renders the dependence universal across moduli. All quantities are computed from the eight fixed-count windows of Table 4.1. No new runs are used.

5.2 Total Decrease of S(M, L)

Table 5.1 — Total relative decrease of S from A (L = 19.96) to P5 (L = 29.93).

M	S(A)	S(P5)	ΔS	Relative decrease
6	0.018154	0.011521	0.006633	36.5%
30	0.148372	0.092723	0.055649	37.5%
210	0.304672	0.226046	0.078626	25.8%
2310	0.418677	0.334435	0.084242	20.1%
30030	0.509198	0.423067	0.086131	16.9%
510510	0.576448	0.494000	0.082448	14.3%

Two observations follow. The absolute decrease grows across the lower part of the modulus ladder and flattens at the largest tested moduli. The relative decrease provides the more comparable scale-free summary. The relative decrease is not strictly

monotone in M: it is 36.5% at M = 6 and 37.5% at M = 30, then falls monotonically to 14.3% at M = 510510. The rate of decrease is not uniform across the modulus ladder.

5.3 Total Decrease of ρ_resid(M, L)

Table 5.2 — Total relative decrease of |ρ_resid| from A to P5.

M	ρ(A)	ρ(P5)	|ρ(P5)/ρ(A)|	Relative decrease
6	−0.029194	−0.018701	0.6406	35.9%
30	−0.025414	−0.017189	0.6764	32.4%
210	−0.020913	−0.014718	0.7037	29.6%
2310	−0.017636	−0.012711	0.7207	27.9%
30030	−0.015026	−0.011071	0.7367	26.3%
510510	−0.013090	−0.009762	0.7457	25.4%

The pattern matches that of S: larger relative decrease at small M, smaller at large M. The contrast is weaker than for S (35.9% versus 25.4%, a ratio of 1.4, compared with 2.6 for S), but the direction is the same.

Both S and ρ_resid therefore share the same qualitative behaviour: they weaken with L, and they weaken faster at small M.

5.4 Descriptive Power-Law Fits

For each M we fit S(M, L) = a · L^(−α) and |ρ_resid(M, L)| = a · L^(−α) to the eight fixed-count points by least squares. The fitted exponents are reported below as descriptive summaries over the observed range, not as claimed asymptotic laws. Section 6 shows that the power-law form is not preferred over alternative two-parameter forms within the range L ∈ [19.96, 29.93].

Table 5.3 — Descriptive power-law fits S(M, L) = a · L^(−α).

M	a	α	R^2
6	0.6143	1.170	0.994
30	5.110	1.177	0.995
210	2.932	0.753	0.997
2310	2.275	0.564	0.998
30030	2.016	0.458	0.999
510510	1.831	0.385	0.999

Table 5.4 — Descriptive power-law fits |ρ_resid(M, L)| = a · L^(−α).

M	a	α	R^2
6	0.8412	1.117	0.989
30	0.4869	0.980	0.995
210	0.2973	0.881	0.998
2310	0.2114	0.824	0.999
30030	0.1556	0.775	0.999
510510	0.1231	0.742	0.999

The α_S ladder is approximately decreasing in M, with a small non-monotonicity at M = 6 and M = 30 (α_S = 1.170 and 1.177 respectively); it declines smoothly thereafter, from 1.170 at M = 6 to 0.385 at M = 510510. The α_ρ ladder declines monotonically from 1.117 to 0.742 over the same range. The two ladders cross between M = 30 and M = 210.

The crossing is an empirical feature of the selected fit and is not assigned a mechanism. No claim is made that the crossing corresponds to any structural property of the modulus ladder.

5.5 Normalised Ratios

To test whether the position dependence is universal across moduli, we normalise each S-curve to its value at the reference window A:

R(M, L) ≡ S(M, L) / S(M, A).

Table 5.5 — Normalised ratios R(M, L).

Window	L	R(6)	R(30)	R(210)	R(2310)	R(30030)	R(510510)
A	19.96	1.000	1.000	1.000	1.000	1.000	1.000
P1	21.30	0.958	0.958	0.973	0.979	0.982	0.985
B	23.13	0.865	0.860	0.905	0.925	0.938	0.948
P2	23.13	0.864	0.859	0.905	0.925	0.938	0.947
P3	25.34	0.773	0.768	0.839	0.876	0.896	0.912
C	27.63	0.688	0.692	0.791	0.839	0.868	0.887
P4	27.63	0.688	0.692	0.791	0.839	0.868	0.887
P5	29.93	0.635	0.625	0.742	0.799	0.831	0.857

No exact collapse is observed. At the smallest displacement (P1, ΔL ≈ 1.33), the six M-curves nearly coincide (R between 0.958 and 0.985). As L increases, the curves separate: at P5, R ranges from 0.625 (M = 30) to 0.857 (M = 510510), a spread of about 1.37 across the modulus ladder.

The position dependence is therefore not described by a single universal shape. At the smallest tested displacement, the normalized values are numerically close across the tested moduli. With only one such displacement, this observation is not interpreted as local universality.

5.6 Summary of Section 5

The following are empirical observations over the fixed-count set, independent of any specific functional form:

- **F1.** For every M, S(M, L) decreases monotonically with L across the eight windows.
- **F2.** For every M, |ρ_resid(M, L)| decreases monotonically with L.
- **F3.** The total relative decrease from A to P5 depends on M: 36.5% at M = 6 versus 14.3% at M = 510510 for S; 35.9% versus 25.4% for ρ_resid.
- **F4.** The fitted power-law exponent α_S(M) is approximately decreasing in M, with a small non-monotonicity at M = 6 and M = 30 (α_S = 1.170 and 1.177 respectively), declining smoothly thereafter; α_ρ(M) is monotone decreasing over the observed ladder. The two sequences cross between M = 30 and M = 210.
- **F5.** No simple normalisation collapses all M-curves onto a single shape.

The following are not established:

- **N1.** A specific mathematical form for S(M, L) or ρ_resid(M, L).
- **N2.** Independently validated values for α_S(M) or α_ρ(M); the fitted exponents are descriptive summaries from a single dataset.
- **N3.** Universality of the L-dependence across M.

Section 6 examines N1 and the identifiability of the functional form.

6. Functional-Form Sensitivity

6.1 Motivation

Section 5 reported descriptive power-law fits with $R^2 \geq 0.989$. High R^2 over a narrow L range does not establish a power law; many smooth functions fit the same data with comparable R^2. This section compares the power-law form against three alternatives on the same data.

6.2 Models Considered

Four two-parameter forms were fit to S(M, L) and ρ_resid(M, L) for each M:

Table 6.1 — Candidate functional forms.

Label	Form	Parameters
power	y = a · L^(−α)	(a, α)
reciprocal	y = a + b / L	(a, b)
log	y = a + b · log(L)	(a, b)
linear	y = a + b · L	(a, b)

All four have two free parameters, so information-criterion comparisons are not affected by differences in model complexity. Fits used nonlinear least squares with

moment-based starting values (paper3_model_comparison.py; sha256 in the data inventory). Two metrics are reported in the main text: the residual sum of squares from the full fit, and the small-sample corrected Akaike Information Criterion AICc = n · ln(RSS/n) + 2k + 2k(k+1)/(n−k−1).

For n = 8 and k = 2, the correction term is 2k(k+1)/(n−k−1) = 2·2·3/5 = 2.4. With only eight windows, AICc rankings are descriptive sensitivity summaries rather than decisive model-selection evidence. Models with ΔAICc < 2 are described as similarly supported within the present candidate set; the term is not used in a strict probabilistic sense.

6.3 Results for S(L), Fixed-Count Set

Table 6.2 — S(L) model comparison, fixed-count set (n = 8 windows per M).

M	Winner	ΔAICc to 2nd	Ranking
6	log	0.65	log > reciprocal > power > linear
30	log	1.44	log > reciprocal > power > linear
210	log	0.28	log > power > reciprocal > linear
2310	power	0.28	power > log > reciprocal > linear
30030	log	0.06	log > power > reciprocal > linear
510510	power	0.47	power > log > reciprocal > linear

Every ΔAICc between best and second-best is below 2. No unique winner emerges among the specific two-parameter forms considered over the present finite L range.

Among the four candidate forms, the linear-in-L model has substantially poorer AICc support for S(M, L) over this dataset, with ΔAICc between 5.35 and 8.17 relative to the leading model in every case (Table A.6). This indicates genuine curvature in S with respect to L over the observed range.

6.4 Results for ρ_resid(L), Fixed-Count Set

Table 6.3 — ρ_resid(L) model comparison, fixed-count set.

M	Winner	ΔAICc to 2nd	Ranking
6	log	7.8	log > linear > reciprocal > power
30	log	2.3	log > linear > power > reciprocal
210	linear	3.1	linear > log > power > reciprocal
2310	linear	5.2	linear > log > power > reciprocal
30030	linear	4.9	linear > log > power > reciprocal
510510	linear	5.8	linear > log > power > reciprocal

For ρ_resid, the picture differs from S. At small M (6, 30), the logarithmic form has the best AICc support, with ΔAICc of 7.8 and 2.3. At large M (210–510510), the linear form

has the best support, with ΔAICc between 3.1 and 5.8. The power-law and reciprocal forms are not preferred for ρ_resid at any M in the fixed-count set.

The apparent change in preferred form with M is recorded as an exploratory observation. With only six values of M, and without an independent validation set, this pattern is suggestive but not confirmed.

6.5 Results on the Combined Window Set

The same comparison was applied to the combined set of 11 windows (adding W1, W2, W3 from protocol v1, whose gap counts vary by a factor of approximately 80.5 (from 10,490,870 in W1 to 844,953,415 in W3)).

Table 6.4 — S(L) model comparison, combined set (n = 11).

M	Winner	ΔAICc to 2nd
6	reciprocal	1.4
30	reciprocal	0.7
210	power	0.6
2310	power	1.9
30030	power	0.7
510510	power	1.7

Table 6.5 — ρ_resid(L) model comparison, combined set.

M	Winner	ΔAICc to 2nd
6	reciprocal	0.4
30	reciprocal	0.02
210	log	2.1
2310	log	2.6
30030	log	1.6
510510	log	0.9

The winner changes when n is no longer fixed. For S(L), the fixed-count set alternates between log and power; the combined set alternates between reciprocal and power. For ρ_resid, the fixed-count set gives log at small M and linear at large M; the combined set gives reciprocal at small M and log at large M.

This inconsistency across datasets is a feature of a dataset whose L-range is too narrow to separate the candidate forms. The change in the leading candidate across the fixed-count and combined datasets reinforces the sensitivity of these model rankings to window design and sample composition.

6.6 Summary of Section 6

Within the observed range L ∈ [19.96, 29.93]:

- No unique winner emerges among the four two-parameter forms considered for S(M, L).
- The linear-in-L form has substantially poorer AICc support for S(M, L) over this dataset.
- For ρ_resid(M, L), the preferred form in the fixed-count set changes with M (log at small M, linear at large M), but the observation is exploratory.
- The identity of the winning form is sensitive to which windows are included in the fit.

6.7 What This Section Does Not Establish

- **N1.** The comparison does not establish that power-law, log, or reciprocal forms are the true functional forms. It establishes only that the logarithmic and power-law forms are similarly supported, while the reciprocal form is less supported at large moduli (M = 510510, 30030).
- **N2.** The comparison does not establish that the true form lies within the four tested. Other smooth forms of comparable complexity could also fit the data over this range.
- **N3.** The comparison does not establish asymptotic behaviour. As L → ∞, the four candidate forms have qualitatively different limits. Extrapolation beyond the observed range is not justified.
- **N4.** The crossing of the α_S and α_ρ sequences reported in Section 5.4 is not explained by this section.

6.8 Recommendations for Future Work

To break the non-identifiability, future experiments should pursue one or more of the following:

(a) Extend the L range. The current set spans L ∈ [19.96, 29.93], a factor of approximately 1.50. A factor of 2 or more would begin to distinguish the four candidates. Windows at 10^14 and 10^15 would extend the L factor to approximately 1.61 and 1.73 respectively; these windows are feasible according to the estimate in §8.4. A factor of 2 would require a window near 2 × 10^17, whose feasibility has not been assessed.

(b) Add additional squarefree CRT-aligned moduli, ideally intermediate primorial products or explicitly documented products of distinct small primes.

(c) Independent sieve verification. Extend the independent comparison to a second high-position window, preferably near 10^13. The internal reproductions reported in Sections 3.5 and 4.5 are encouraging but do not exclude a systematic error shared by all runs.

(d) Broader detrending forms. A systematic study using local polynomial regression, splines, or detrending on log log p would test whether the ABCD decomposition is robust to broader families of detrending choices.

(e) Joint bootstrap across windows. A joint bootstrap would provide confidence intervals for the fitted exponents and for the crossing point of the α-ladder. This is a

methodological improvement, not a new empirical question.

7. Claims and Non-Claims

7.1 Purpose

Sections 3–6 established a set of empirical facts about the phase-decomposed covariance of prime gaps, together with a demonstrated limitation: the functional form of the position dependence is not identified by the available data. This section states explicitly what the paper claims and what it does not. The separation matters because the observed trends are strong, systematic, and reproducible, yet they are confined to a narrow range of L and their interpretation requires care.

7.2 Claims We Make

C1. Falsification of T6. The preregistered prediction T6 (that S(M) at higher windows would be greater than or equal to S(M) at lower windows, for the same M) is falsified. S(M) decreases systematically from W1 to W3 for every M tested.

- Support: Table 3.2; all adjacent window pairs decrease, and all 18 pairwise window–modulus comparisons decrease
- Scope: W1 = $[10^8, 3\times10^8)$, W2 = $[10^9, 3\times10^9)$, W3 = $[10^{10}, 3\times10^{10})$

C2. Robustness to linear detrending. The decrease of S(M) is not an artifact of the linear form of the global detrending. Re-running the analysis with a quadratic detrending $g \approx a + b\cdot z + c\cdot z^2$ leaves S unchanged to six decimal places in all 19 cases (6 in W1, 6 in W2, 7 in W3). See Table A.7 for the per-window, per-modulus comparison.

- Scope: Robust to one specific smooth-curvature alternative; not a proof of invariance under arbitrary detrending choices.

C3. Position dependence with gap count held approximately fixed. Across eight windows at $n \approx 10^8 \pm 5.1\%$, S(M) and |ρ_resid(M)| decrease monotonically with L = log p for every M tested. The total relative decrease is 14–37% for S and 25–36% for ρ_resid, depending on M.

- Scope: $n \approx 10^8$; $L \in [19.96, 29.93]$; six moduli.

C4. Strong disfavouring of the sample-size hypothesis. The decrease of S(M) is not attributable to a pure sample-size effect, because n varies by less than 5.1% across the eight fixed-count windows while S varies by 14–37%.

- Scope: Empirical, not causal; see §4.7 for caveats on window width in x and on the proxy role of L.

C5. Sensitivity of functional form. For S(M, L), the logarithmic form is within $\Delta\text{AICc} < 2$ of the leading model at every modulus, the power-law form at five of six, and the reciprocal form at three of six. The linear-in-L form is substantially poorer, with ΔAICc between 5.35 and 8.17 relative to the leading model. For ρ_resid(M, L),

the preferred form in the fixed-count set changes with M, but the observation is exploratory.

- Scope: Four two-parameter forms over L ∈ [19.96, 29.93]; n = 8 windows per M.

C6. Persistence of negative residual correlation. For every M in every window analysed, ρ_resid < 0, and every within-window block-bootstrap 95% confidence interval excludes zero.

- Scope: 48/48 values negative in the fixed-count set; all 19 within-window bootstrap intervals in the W-set are strictly negative (6 in W1, 6 in W2, 7 in W3). Intervals are marginal within-window; they are not treated as independent tests.

C7. Internal consistency and reproducibility. The ABCD decomposition closes to machine precision in every window (max |A+X+Y+D − Cov| ≤ 7.1×10^{-14}). The variance split holds to machine precision. The window W3 reproduces Paper 2 exactly at M = 9699690 (S = 60.4431%, ρ_resid = −0.010306). Two pairs of fixed-count windows at duplicated positions (B/P2, C/P4) differ by at most 3.7 × 10^-4 in S (at M = 510510) and 1.7 × 10^-5 in ρ_resid. These differences exceed the last printed digit for the larger-modulus S values but remain ~200× smaller than the total A-to-P5 decrease reported in Section 5.

7.3 Claims We Do Not Make

N1. No asymptotic scaling law. We do not claim that S(M, L) ~ L^(−α(M)) or any other specific form is the correct asymptotic behaviour. The fitted exponents reported in Section 5.4 are descriptive summaries over the observed L range. As L → ∞, the four candidate models of Section 6 have qualitatively different limits, and the data cannot distinguish them.

N2. No validated exponent ladder. The fitted exponent α_S(M) is approximately decreasing in M, with a small non-monotonicity at M = 6 and M = 30 (α_S = 1.170 and 1.177 respectively), while α_ρ(M) appears monotone decreasing over the observed modulus ladder, but these are descriptive quantities from a single dataset and have not been independently validated. The crossing of the two sequences between M = 30 and M = 210 is an empirical feature of the selected fit, not a confirmed law.

N3. No universality of the L-dependence. The normalised ratios R(M, L) = S(M, L)/S(M, A) do not collapse onto a single curve. The shape of S(M, L) depends on M, and no simple rescaling eliminates this dependence over the observed range.

N4. No comparison with analytic number theory. We do not derive our results from Montgomery–Odlyzko pair correlation, the Goldston–Pintz–Yıldırım small-gap results, the Maynard bounded-gap theorem, or the Lemke Oliver–Soundararajan residue-pair biases. The Lemke Oliver–Soundararajan study of consecutive-prime residue-pair biases [9] is the closest contextual framework to our residue-class conditioning, but to our knowledge, it does not provide a quantitative prediction for the specific non-D covariance share studied here. We do not claim a comparison that the present data cannot support.

N5. No connection to zeta zeros. We make no claim about the distribution of non-trivial zeros of the Riemann zeta function. The prime-gap data analysed here are

strictly finite-range and empirical.

N6. No claim beyond the observed range. No claim is made about the sign of ρ_resid, the form of S(M, L), or the rate of decay at positions beyond the windows examined here.

N7. No test of Cramér's model. The present paper does not test Cramér's heuristic model directly, nor does it evaluate the model's ability to reproduce the observed position dependence.

N8. No statistical comparison with Papers 1 and 2 beyond reproduction. Paper 1 established persistent negative lag-one dependence up to 10^9; Paper 2 established the ABCD decomposition at a single window. Paper 3 extends these results to new windows. Beyond the direct reproduction of Paper 2's central number (C7), no statistical comparison between papers is offered.

7.4 Relation to Papers 1 and 2

Paper 1 [1] (Zenodo DOI 10.5281/zenodo.23162269) established persistent negative lag-one dependence in locally detrended prime-gap windows up to N = 10^9. The present work extends the analysis to higher positions (up to 10^{13}) and to a larger family of moduli, and confirms that the negative residual correlation persists across the extended range (C6).

Paper 2 [2] (Zenodo DOI 10.5281/zenodo.23222536) introduced the ABCD decomposition and applied it to the single window [10^{10}, 3×10^{10}) with 844,953,415 gaps. The central number of Paper 2 — S = 60.4% at M = 9699690 — is reproduced exactly in W3 of the present work (C7). This identity verifies that the pipeline of Paper 3 is a faithful extension of Paper 2's code.

Paper 3 does not supersede Papers 1 and 2; it extends them.

7.5 Methodological Caveats

(a) Detrending. We detrend with a global polynomial in log p of degree at most two. Other smooth forms (splines, local regression) may yield different residual distributions. The quadratic diagnostic (C2) addresses one specific alternative to linear detrending and does not establish invariance under arbitrary detrending choices.

(b) Sieve and base primes. Gap enumeration uses an odd-only segmented sieve with base primes up to ⌈√HI⌉ + 100. The small-scale validations (10^6, 10^7) reproduce known prime counts exactly. An independent enumeration check was completed at 10^{12} (Appendix B), and a numerical-stability check of the OLS detrending is reported in Appendix C. A second independent high-position enumeration check near 10^{13} remains desirable.

(c) Bootstrap. The block bootstrap (40 blocks, 1000 replicates, seed base 20261006 + k·1000) provides confidence intervals for ρ_resid within a single window. It does not provide confidence intervals for the fitted exponents α(M), which would require a joint bootstrap across windows. The α values are therefore reported as descriptive only, consistent with C5 and N1.

(d) Modulus eligibility. Modulus eligibility was determined dynamically by the minimum class count over all residues in the window, with a modulus excluded when its smallest non-empty class count is below 50 and classes between 50 and 99 flagged diagnostic. This rule differs from the fixed seven-modulus list used in Paper 2.

7.6 Summary

Paper 3 establishes seven claims (C1–C7) and explicitly declines eight (N1–N8). The claims concern falsification of a preregistered prediction, a fixed-count diagnostic that strongly disfavours a pure sample-size explanation, an empirical description of the position dependence, a sensitivity analysis of the functional form, and internal consistency checks. The paper does not advance a theoretical model, does not connect to analytic number theory, and does not claim asymptotic behaviour.

8. Outlook and Open Questions

8.1 The Central Open Question

The empirical result of this paper can be stated in one sentence: the phase-decomposed lag-one covariance of prime gaps weakens systematically with window position L = log p, at approximately fixed gap count, across the range L ∈ [19.96, 29.93]. No single two-parameter functional form is uniquely identified by the data.

This raises a question the paper does not answer: what is the mechanism?

The question can be phrased in several equivalent ways:

- Why does the non-D covariance share S(M) decrease as the window moves to higher position?
- Why does the residual correlation ρ_resid — the part not captured by any finite-modulus phase structure — weaken in the same direction?
- Is the weakening a property of the prime sequence itself, or of the estimation of residue-class means over a finite window?

Several explanations have been examined and, in some cases, disfavoured by the analyses; the mechanism remains open.

8.2 What Has Been Disfavoured

(a) Sample size. The fixed-count set holds n within 5.1%, while S varies by 14–37%. The design removes the most direct sample-size confounding; a pure sample-size explanation is strongly disfavoured.

(b) Linear detrending artifact. The quadratic diagnostic of Section 2.4, performed on W1–W3 only, shows that the decrease survives this specific smooth-curvature alternative on those windows.

(c) Position-calibration artifact. No evidence of a position-calibration artifact appears in the duplicated-window check: the paired estimates differ by at most 3.7 × 10^-4 in S (at M = 510510) and 1.7 × 10^-5 in ρ_resid. These differences exceed the

last printed digit for the larger-modulus S values but remain ~200× smaller than the total A-to-P5 decrease reported in Section 5. This does not exclude a systematic error shared by all runs.

Beyond these three exclusions, the cause of the position dependence remains open.

8.3 Candidate Directions

The following directions are recorded as possible explanations. None is advanced as a claim.

(i) Finite-range effect. If the underlying phase structure of prime gaps is position-independent, the observed weakening might reflect the finiteness of the window: as L grows, the per-class sample mean becomes a poorer estimator of the underlying structure, and S declines without any change in the true structure. A finite-range explanation could produce curvature, slowing, saturation, or a crossover outside the observed range. The present data do not distinguish these possibilities from continued decay.

(ii) Genuine structural drift. Alternatively, the weakening could reflect a change in the distribution of consecutive gaps as the prime axis is traversed, with residue-class structure becoming less informative at high position. Under this hypothesis, the weakening continues beyond the observed range.

The present data cannot distinguish (i) from (ii). This distinction is the principal open question raised by the paper.

(iii) Residue-class sparsity interaction. The decrease in S is greater at small M and smaller at large M. If the mechanism operated through the number of classes per modulus φ(M), a different pattern might be expected. The observed pattern does not point to a simple class-count mechanism.

(iv) Estimator-distribution interaction. The per-residue-class means μ[r] are estimated from the same gap sequence as the residuals. If the estimation error decreases with sample size more slowly than the true phase signal, S may appear to decline. This hypothesis is related to (i) but distinct: it concerns the interaction between the estimator and the underlying distribution.

No mechanism is identified. The four directions are listed to guide future work, not to select among them.

8.4 Concrete Experimental Paths

The following experiments would advance the open questions and are feasible within the current computational setup.

(a) Extend the L range. Current range: $L \in [19.96, 29.93]$, factor 1.50. Windows at 10^{14} and 10^{15} would extend this to factors of approximately 1.61 and 1.73 respectively; a factor of 2 would require a window near 2×10^{17}. With the current sieve, a window at 10^{14} with $n \approx 10^8$ gaps requires approximately 10–15 minutes; at 10^{15}, approximately 1–2 hours. These times are estimates, not measurements. This is the most direct route to test whether the weakening slows or continues.

(b) Add additional squarefree CRT-aligned moduli, ideally intermediate primorial products or explicitly documented products of distinct small primes.

(c) Independent sieve verification. Extend the independent comparison to a second high-position window, preferably near 10^{13}. Compare counts, first and last primes, and gap sums. This would eliminate residual concern about systematic error at large position.

(d) Broader detrending forms. A systematic study using local polynomial regression, splines, or detrending on log log p would test whether the ABCD decomposition is robust to broader families of detrending choices.

(e) Joint bootstrap across windows. A joint bootstrap would provide confidence intervals for the fitted exponents and for the crossing point of the α-ladder. This is a methodological improvement, not a new empirical question.

8.5 What This Paper Does Not Attempt

Paper 3 is an empirical report. It does not attempt a theoretical model of prime-gap phase structure, a connection to analytic number theory, a prediction for other arithmetic sequences, or computational optimization of the current pipeline. These omissions are deliberate.

8.6 A Note on Method

This project began with a preregistered protocol (v1) containing seven predictions. One of them, T6, failed. The failure drove the subsequent investigation reported here.

A falsified prediction is data, not embarrassment. The choice to freeze a prediction in advance and to report its failure without modification is what makes the surprising direction of the effect credible. Had T6 been formulated after seeing the data, the observed decrease of S with position would carry much less evidential weight.

This is recorded as a description of how the present results were obtained, not as a methodological prescription.

8.7 Closing

The phase-decomposed lag-one covariance of prime gaps decreases systematically with window position $L = \log p$, at approximately fixed gap count, across the range $L \in [19.96, 29.93]$. The decrease is monotone in every modulus tested, robust to the choice of global detrending form, reproduced at duplicated positions, and not attributable to a pure sample-size effect.

The decrease is not described by any single two-parameter functional form that the available data identify. Logarithmic and power-law descriptions are similarly supported, while the reciprocal form is less supported at large moduli ($M = 510510$, 30030); the linear-in-L form is substantially disfavored.

Whether the weakening slows at higher position or continues indefinitely is the principal open question. The answer lies beyond the range accessible to the present study.

We offer the result as an empirical finding, not a theoretical claim.

8.8 Acknowledgments

This work was carried out with extensive assistance from large-language-model systems at every stage of the project:

- **DeepSeek** contributed the initial theoretical framing of the ABCD decomposition, the design of the preregistered protocol v1, the analysis of the T6 falsification, and the fixed-count diagnostic that separates gap count from window position.
- **Claude, accessed via Perplexity**, provided multiple rounds of critical methodological review. These reviews identified the algebraic redundancy between the lag-one covariance, the residual correlation, and the residual variance; the non-identifiability of power-law, logarithmic, and reciprocal descriptions over the observed narrow range of L; the nested structure of the paired windows B ⊂ P2 and C ⊃ P4; and numerous textual and statistical corrections throughout the manuscript.
- **OpenCode** served as the local execution platform. Within it, three successive models carried out the computational work: **Big Pickle**, **Long Cat**, and finally **Muse Spark**. Together they executed all numerical runs, generated the CSV outputs, performed the independent sieve verification at 10^12 against primesieve 12.16, ran the numerical-stability check with the centered-variable implementation, and produced the LaTeX/PDF build and the reproducibility package.
- **Kimi** and **GPT 6.1 Sol** provided additional critical reviews that identified further numerical and conceptual inconsistencies, including the numerical underdetermination of the OLS slope in high-position windows, the AICc comparison error, the significant-digits claim in the paired-window check, and several reference and citation issues.

The human author (R.B.) is solely responsible for the scientific claims, the design decisions, the choice of methods, and any errors that remain.

A full reproducibility package — source code, CSV outputs, versioned protocols, data inventory with SHA-256 hashes, and the verification logs — is deposited alongside this manuscript on Zenodo.

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Appendix A — Extended Tables

A.1 Descriptive Power-Law Fits for S(M, L)

Referenced from Section 5.4. Fitted exponents α_S(M) are descriptive summaries over the observed range L ∈ [19.96, 29.93]; they are not claimed as asymptotic laws.

Table A.1 — Descriptive power-law fits S(M, L) = a · L^(−α).

M	a	α	R^2
6	0.6143	1.170	0.994
30	5.110	1.177	0.995
210	2.932	0.753	0.997
2310	2.275	0.564	0.998
30030	2.016	0.458	0.999
510510	1.831	0.385	0.999

A.2 Descriptive Power-Law Fits for |ρ_resid(M, L)|

Referenced from Section 5.4. Same caveat as A.1.

Table A.2 — Descriptive power-law fits |ρ_resid(M, L)| = a · L^(−α).

M	a	α	R^2
6	0.8412	1.117	0.989
30	0.4869	0.980	0.995
210	0.2973	0.881	0.998
2310	0.2114	0.824	0.999
30030	0.1556	0.775	0.999
510510	0.1231	0.742	0.999

A.3 Normalised Ratios R(M, L)

Referenced from Section 5.5. The normalisation R(M, L) = S(M, L) / S(M, A) tests whether the position dependence is universal across moduli. No exact collapse is observed.

Table A.3 — Normalised ratios R(M, L).

Window	L	R(6)	R(30)	R(210)	R(2310)	R(30030)	R(510510)
A	19.96	1.000	1.000	1.000	1.000	1.000	1.000
P1	21.30	0.958	0.958	0.973	0.979	0.982	0.985
B	23.13	0.865	0.860	0.905	0.925	0.938	0.948
P2	23.13	0.864	0.859	0.905	0.925	0.938	0.947
P3	25.34	0.773	0.768	0.839	0.876	0.896	0.912
C	27.63	0.688	0.692	0.791	0.839	0.868	0.887
P4	27.63	0.688	0.692	0.791	0.839	0.868	0.887
P5	29.93	0.635	0.625	0.742	0.799	0.831	0.857

A.4 Model Comparison on the Combined Window Set

Referenced from Section 6.5. The combined set includes W1, W2, W3 from protocol v1 whose gap counts vary by a factor of approximately 80.5 (from 10,490,870 in W1 to 844,953,415 in W3).

Table A.4 — S(L) model comparison, combined set (n = 11).

M	Winner	ΔAICc to 2nd
6	reciprocal	1.4
30	reciprocal	0.7
210	power	0.6
2310	power	1.9
30030	power	0.7
510510	power	1.7

Table A.5 — ρ_resid(L) model comparison, combined set.

M	Winner	ΔAICc to 2nd
6	reciprocal	0.4
30	reciprocal	0.02
210	log	2.1
2310	log	2.6
30030	log	1.6
510510	log	0.9

A.5 Full ΔAICc Table

Referenced from Section 6.3. The main text of Section 6.3 reports only the difference between the leading and the second-ranked model at each modulus. Table A.6 lists the full ΔAICc for all four candidate models at each modulus, extracted from model_comparison_output.txt (PRIMARY fixed-n set, quantity S). The comparison script labels the reciprocal model "inverse"; the paper uses "reciprocal" for the same y = a + b / L form.

Table A.6 — Full ΔAICc, S(L) on the fixed-n set (n = 8).

M	power	reciprocal	log	linear
6	1.753	0.651	0.000	6.804
30	2.933	1.440	0.000	8.172
210	0.280	1.856	0.000	7.168
2310	0.000	2.549	0.281	7.388
30030	0.060	2.591	0.000	5.350
510510	0.000	2.940	0.465	7.030

A.6 Linear vs Quadratic Detrending

Referenced from C2. Table A.7 compares S_linear (from the variable-count CSVs of Section 3) and S_quad (from the quadratic-diagnostic CSVs) for each window and modulus. All differences are 0 to the six decimal places of the CSV extraction precision, hence below 10^-6.

Table A.7 — S_linear vs S_quad per window and modulus.

Window	M	S_linear	S_quad	Difference
W1	6	0.019670	0.019670	0
W1	30	0.162635	0.162635	0
W1	210	0.323706	0.323706	0
W1	2310	0.437143	0.437143	0
W1	30030	0.525091	0.525091	0
W1	510510	0.592241	0.592241	0
W2	6	0.017458	0.017458	0
W2	30	0.142455	0.142455	0
W2	210	0.296759	0.296759	0
W2	2310	0.410226	0.410226	0
W2	30030	0.500907	0.500907	0
W2	510510	0.568517	0.568517	0
W3	6	0.015317	0.015317	0
W3	30	0.125040	0.125040	0
W3	210	0.272725	0.272725	0
W3	2310	0.385438	0.385438	0
W3	30030	0.475868	0.475868	0
W3	510510	0.544106	0.544106	0
W3	9699690	0.604431	0.604431	0

Appendix B — Independent Sieve Verification

B.1 Purpose

The scientific claims of this paper depend on an accurate enumeration of prime gaps at positions up to 10^13. This appendix documents an independent cross-check of the enumeration at position 10^12 against a separate, widely-used sieve implementation.

B.2 Method

Prime enumeration in test_multiwindow_v1.exe was compared against primesieve, version 12.16, an independently developed library for prime generation. The comparison window was

[10^12, 10^12 + 10^7),

chosen to be small enough to run quickly (~2 seconds for our binary) and large enough to exercise the high-position arithmetic pathways.

For each window, five quantities were compared:

1. Total count of primes in the window (n_gaps).
2. First prime in the window.
3. Last opening prime in the window.
4. Closing prime (the smallest prime greater than HI).

5. Sum of gaps from first to closing.

Additionally, SHA-256 digests of the first 1000 and last 1000 primes were compared.

B.3 Results

Quantity	primesieve 12.16	test_multiwindow_v1	Verdict
n_gaps	361,726	361,726	MATCH
first_prime	1,000,000,000,039	1,000,000,000,039	MATCH
last_open	1,000,009,999,981	1,000,009,999,981	MATCH
closing_prime	1,000,010,000,023	1,000,010,000,023	MATCH
sum_of_gaps	9,999,984	9,999,984	MATCH
sha256 first 1000	a3cf8071…	a3cf8071…	MATCH
sha256 last 1000	070bfb13…	070bfb13…	MATCH

All quantities agree exactly. No discrepancy was observed.

B.4 Notes on the verification script

The verification script (sieve_verify.py) invokes primesieve.exe directly through Python's subprocess module, rather than through the Python wrapper. The executable path is platform-dependent and must be adjusted by the reader to run the script on a different system. On the author's system the path was of the form

```
<drive>:\<user-directory>\primesieve-<version>\primesieve.exe
```

The script is otherwise platform-independent in logic.

B.5 Technical caveat

The verification window is narrow: the standard deviation of log p is only about 3 × 10^-6 (the range is about 10^-5), and the OLS detrending within this window is therefore statistically under-determined. The S(M) and ρ_resid(M) values from this window are not used as data points anywhere in the paper; the window serves solely for sieve verification, which does not depend on detrending.

In the eight fixed-count windows used in Sections 4–6, the raw variance of log p ranges from approximately 0.54 (window A) to approximately 7.5 × 10^-9 (window P5). The OLS slope β in the highest-position windows is correspondingly imprecise. A dedicated numerical-stability check (Appendix C) confirms that the reported S(M) and ρ_resid(M) are insensitive to this: an independent implementation using a centered log variable (z = log1p((p − p_mid)/p_mid)) reproduces both quantities to six decimal places at every modulus and in the four narrow windows tested (P3, P4, P5, C).

B.6 Artifacts

Verification artifacts are recorded in paper3_data_inventory.txt:

sieve_verify.py sha256=22a4261d…
sieve_verify_1e12.log sha256=1f5e18ac…
phase_decomp_1000000000000_1000010000000.csv sha256=785be809…

B.7 Statement

We conclude that the enumeration agrees with primesieve 12.16 on the specified verification window near 10^12. No discrepancy was observed in the compared quantities.

Appendix C — Numerical Stability of the OLS Detrending

C.1 Purpose

The eight fixed-count windows span prime locations from 10^8 to 10^13. As the position increases, the log p range within a window narrows, and the OLS slope β becomes statistically imprecise. This appendix reports a direct check that the reported quantities S(M) and ρ_resid(M) do not depend on this imprecision.

C.2 Method

The detrending in the main pipeline fits g_n ≈ α + β · log p_n by ordinary least squares over the raw log p values. In the numerical-stability check, we replace this with an algebraically equivalent fit on a centered and scaled variable:

p_mid = (LO + HI) / 2 z_n = log1p((p_n − p_mid) / p_mid) g_n ≈ α’ + β’ · z_n

The two forms are equivalent: β = β’, and the residuals e_n are identical up to floating-point rounding. The centered form avoids subtracting two large logarithms and preserves precision at position 10^13.

We implemented the centered form in a separate source file (test_multiwindow_v2.cpp; sha256 in data inventory) and re-ran four narrow windows (P3, P4, P5, and C) with both pipelines. For each window, S(M) and ρ_resid(M) were extracted from the resulting CSVs and compared.

C.3 Results

Table C.1 — Numerical stability of S and ρ.

Window	Var(log p)	β_v1	β_v2	se(β_v2)	max	ΔS
P3	5.2×10^-5	1.0729	1.0729	0.313	0	0
P4	6.4×10^-7	1.7601	1.7601	3.116	0	0
P5	7.5×10^-9	−2.3099	−2.3098	31.363	0	0
C	6.4×10^-7	1.7317	1.7317	3.108	0	0

All differences max|ΔS| and max|Δρ| are 0.00e+00 to six decimal places (the extraction precision of the CSV files). The largest difference in β between the two pipelines

occurs in P5 and equals 7.1 × 10^-5, which is ~2.3 × 10^-6 of the OLS standard error (se ≈ 31.4).

C.4 Interpretation

Two facts follow:

1. The OLS slope β is statistically under-determined in the highest-position windows. In P5 the standard error of β is ~31, so the fitted value β = −2.31 lies within 0.1 σ of the expected value 1. In P4 and C the standard errors are 3.1 and 3.1 respectively; in P3 it is 0.31. The fitted slopes lie within 0.3 σ of 1 in all four windows.
2. Nevertheless, the quantities S(M) and ρ_resid(M) — which are the reported empirical results — are numerically insensitive to this imprecision. The residuals differ between the two pipelines by at most |β − 1|·Δlog p ≈ 10^-3 in P5, compared with a residual standard deviation of ~27. The fractional effect is < 5 × 10^-5 and is therefore invisible at the reported precision.

C.5 Statement

The four narrow windows examined — P3, P4, P5, and C — do not carry any detectable numerical bias from the OLS detrending. The reported S(M) and ρ_resid(M) values are reproduced exactly by an independent centered-variable implementation. The narrowness of these windows affects the identifiability of β, not the empirical quantities of interest.